UK Space-Based Solar Power Cost-Condition Assessment - v2.0

Report subtitle: Delivered-grid LCOE thresholds under explicit technical and financial assumptions

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Repository: https://github.com/WenyuGao1/space-based-solar-power-lcoe-model

Interactive model: https://wenyugao1.github.io/space-based-solar-power-lcoe-model/html/

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Technical summary

This standalone report asks what a 2 GW grid-delivered UK space-based solar power system could cost under explicit, adjustable and auditable assumptions, and which variables most change that result. The reference scenario produces a conditional DCF LCOE of **£86.03/MWh** in 2024 real GBP. It reconciles discounted lifecycle cost of **£13.10bn** with **152.32 million MWh** of discounted delivered energy.

- The reference case assumes 2 GW grid-delivered AC, 14.98% end-to-end efficiency, 10.00 million kg (10,000 tonnes) of orbital hardware and 118 equivalent launches.
- Discounted initial construction contributes about **£76.86/MWh** and dominates the £86.03/MWh result.
- Within their separate explored ranges, favourable bounds for system specific mass, launch cost to staging orbit and the real discount rate reduce LCOE to **£31.3, £49.5 and £56.9/MWh**, respectively. These are boundary sensitivities, not a jointly achievable design.
- UK and NASA studies likewise show strong dependence on architecture, launch, finance and combined favourable assumptions. Different system boundaries mean their cost ranges provide context and consistency checks, not validation of this model's £86.03/MWh result.

Research question, scope and core definitions

The research question is: without presenting a scenario as a forecast, which technical, deployment, lifecycle and financing conditions move SBSP into this study's £80-120/MWh discussion region? The £80, £100, £120 and £150/MWh lines are analytical screens, not official UK targets.

The system boundary works upstream from 2 GW AC at the grid connection point to incident solar power. Launch is priced per kg and covers staging orbit only. The staging service boundary is LEO; the final operational orbit is architecture-dependent high Earth orbit. The transfer proxy must cover the transfer vehicle, propellant, refuelling missions, operations and payload-performance penalty.

Core definition or assumption	Reference case	Interpretation
Delivered capacity	2.00 GW AC	Rated AC power at the grid connection point
Price basis	2024 real GBP	All model financial inputs and outputs
Delivered capacity factor	90%	Combined proxy for availability, outages and operating constraints
Operating life	30 years	Full operating years after commissioning
Real discount rate	6.5%	Discounted from start of construction at t=0
System specific mass	5.0 kg/kW-delivered	Complete orbital hardware per kW delivered to the grid
Launch cost	£500/kg	To staging orbit only

The stage-resolved chain required for 2 GW grid delivery

Stage	Rated power
Incident solar power	13.35 GW
Space DC-bus output	4.67 GW
Emitted RF power	3.27 GW
RF power incident on rectenna	3.21 GW
Rectenna DC output	2.08 GW
Grid-delivered AC power	2.00 GW

Computed end-to-end efficiency: **14.9822%**.

Discounted-cash-flow and cost boundaries

The headline metric is LCOE = [sum_t C_t / (1+r)^t] / [sum_t E_t / (1+r)^t]. The valuation base is start of construction at t=0. Four equal construction shares are spent before commissioning, and the first operating-year cash flow is at t=5. Operating life is 30 years; first-year delivered energy is 15.77 TWh and lifetime-average annual delivered energy is 14.68 TWh. The simple CRF reconciliation is £71.02/MWh and is secondary because it does not reproduce the full construction timing.

Orbital mass equals delivered capacity (kW) x kg/kW-delivered; it is not divided by chain efficiency again. Space generation, RF transmission, rectenna and grid costs are assigned to their own rated-power denominators so that the same power or cost is not counted twice. Space replacement includes related hardware, launch, transfer and assembly; ground replacement includes rectenna and grid only. Fixed O&M excludes contingency, initial launch, transfer and assembly.

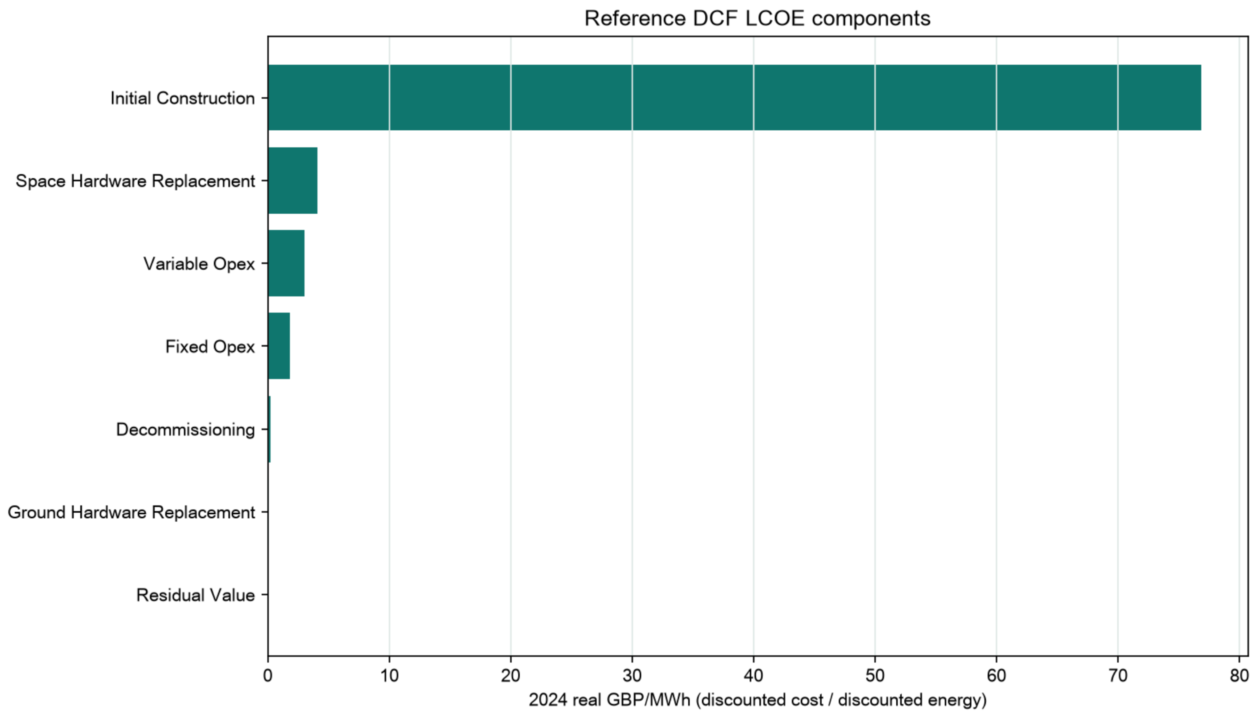
The default construction profile is [0.25, 0.25, 0.25, 0.25] and sums to 100%. Programme contingency is applied once to initial CAPEX. Terminal decommissioning is a cost; residual value is credited at the end of operating life.

Upfront construction dominates the reference cost

CAPEX component	2024 real GBP
Launch to staging orbit	£5.00bn
In-orbit assembly and deployment	£2.00bn
Space-generation hardware	£1.87bn
Transfer to operational orbit	£1.00bn
RF transmitter hardware	£0.33bn
Rectenna	£0.30bn
Grid connection	£0.20bn
Programme contingency	£2.14bn
Total initial CAPEX	£12.84bn

Lifecycle component	Discounted present value	LCOE contribution
Initial construction	£11.71bn	£76.86/MWh
Space-hardware replacement	£0.62bn	£4.08/MWh
Variable O&M	£0.46bn	£3.00/MWh
Fixed O&M	£0.27bn	£1.80/MWh
Decommissioning	£0.03bn	£0.20/MWh
Ground-hardware replacement	£0.02bn	£0.10/MWh
Residual value	£0.00bn	£0.00/MWh

Reference lifecycle LCOE components

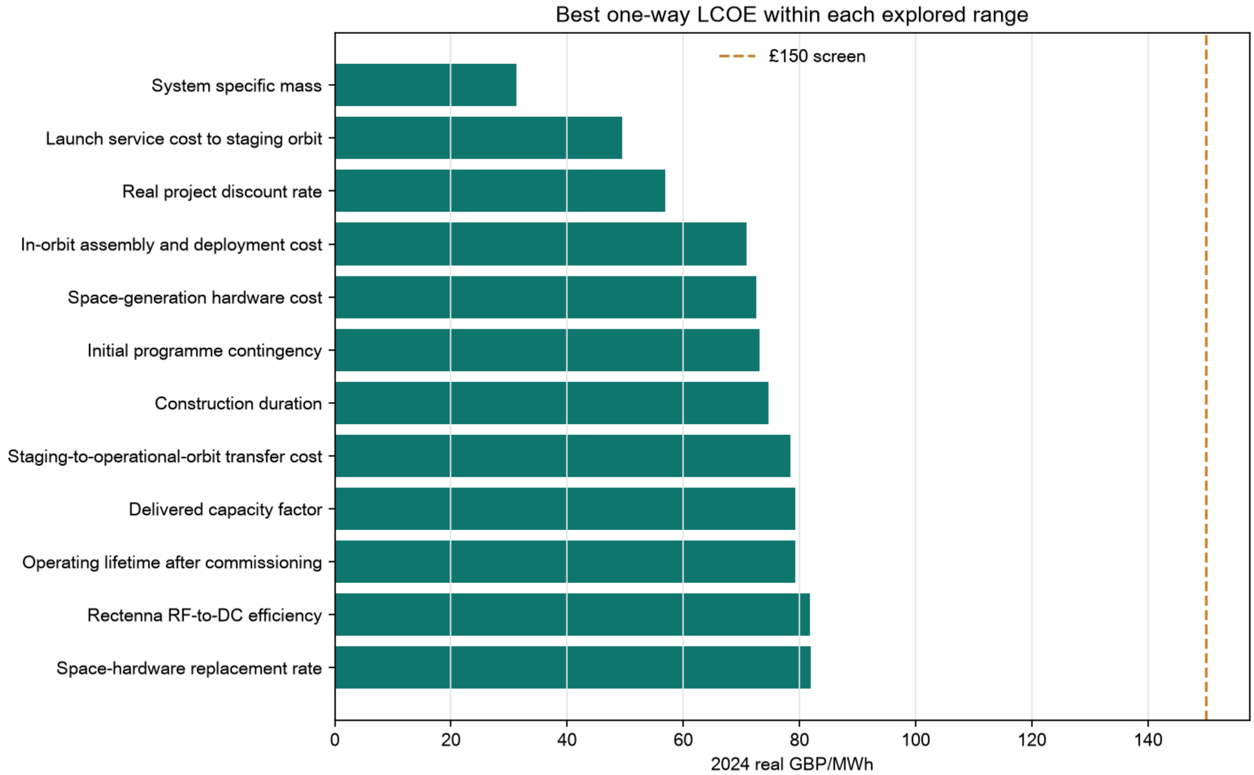


Discounted initial construction contributes **£76.86/MWh**, so the reference result is particularly exposed to upfront CAPEX, construction timing and finance. Other lifecycle items remain material but do not change the predominantly capital-intensive cost structure.

Specific mass, launch and finance are the strongest one-way drivers

Parameter	LCOE at favourable bound	Reduction from reference
System specific mass	£31.3/MWh	£54.7/MWh
Launch service cost to staging orbit	£49.5/MWh	£36.5/MWh
Real project discount rate	£56.9/MWh	£29.1/MWh
In-orbit assembly and deployment cost	£70.8/MWh	£15.2/MWh
Space-generation hardware cost	£72.5/MWh	£13.5/MWh
Initial programme contingency	£73.2/MWh	£12.8/MWh
Construction duration	£74.7/MWh	£11.4/MWh
Staging-to-operational-orbit transfer cost	£78.4/MWh	£7.6/MWh

One-way LCOE at favourable bounds



Each bar moves one parameter to its favourable explored bound while every other input remains at reference. A lower bar indicates a stronger one-way effect, not a higher probability of achievement.

One-way thresholds quantify entry into lower-cost regions

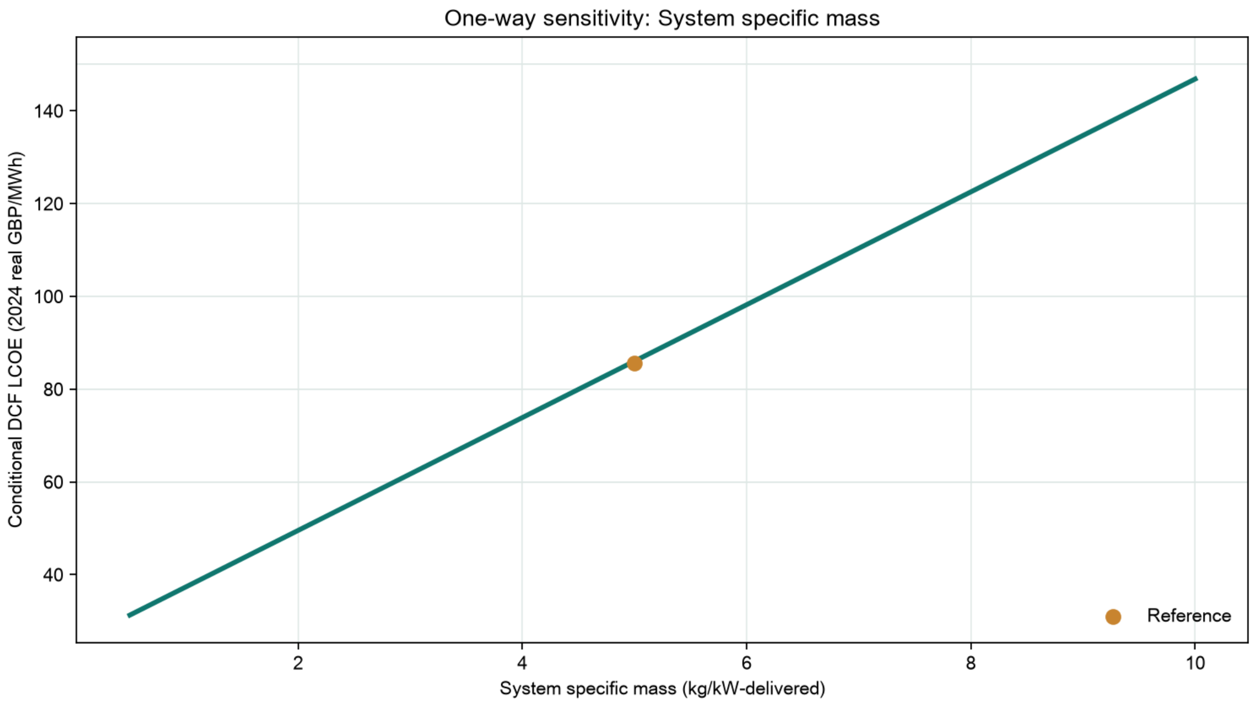
The table reports the most decision-useful one-way thresholds. Each row changes one parameter while all other inputs remain at their reference values. It answers 'how far would this input alone need to move?' without claiming that the condition is achievable on a schedule.

Target LCOE	One-way parameter	Condition that reaches target
£80/MWh	System specific mass	at or below 4.5 kg/kW-delivered
£80/MWh	Launch service cost to staging orbit	at or below £421/kg to staging orbit
£80/MWh	Real project discount rate	at or below 5.85%
£80/MWh	Space-generation hardware cost	at or below £0.244/W-DC
£80/MWh	Construction duration	at or below 1 year
£60/MWh	System specific mass	at or below 2.86 kg/kW-delivered
£60/MWh	Launch service cost to staging orbit	at or below £158/kg to staging orbit
£60/MWh	Real project discount rate	at or below 3.42%

Target	Equal-fraction movement	Computed chain efficiency
£150/MWh	0.0%	14.98%
£120/MWh	0.0%	14.98%
£100/MWh	0.0%	14.98%
£80/MWh	2.9%	15.36%
£60/MWh	12.5%	16.64%

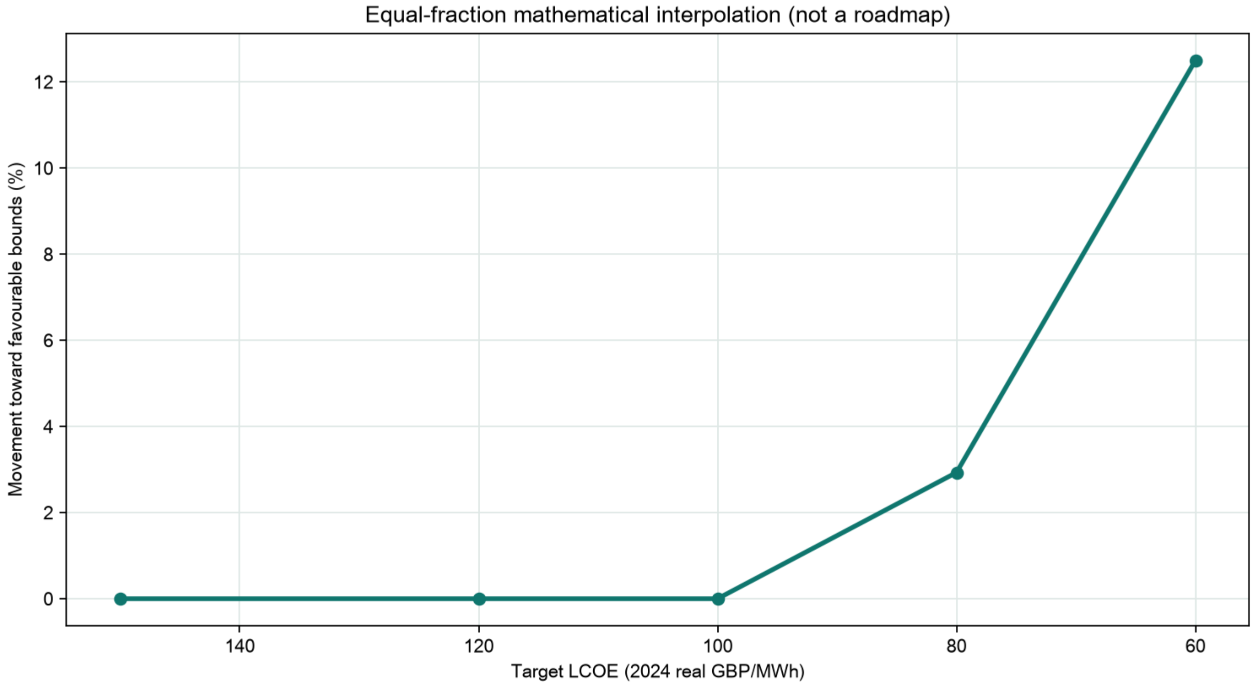
The reference case is already below £150, £120 and £100/MWh, so those targets show 0.0% movement in the equal-fraction table. The £80 and £60/MWh cases interpolate all parameters towards favourable bounds. This frontier is not a readiness score, probability, forecast, schedule or engineering roadmap.

System specific-mass sensitivity



Specific mass affects launch, orbit transfer, in-orbit assembly and subsequent space replacement together; it is not equivalent to reducing a single hardware purchase cost. The plotted linearity follows the current per-kg boundary. A real mission would also face discrete launch counts, structural constraints and vehicle-integration limits.

Equal-fraction combined interpolation



Relationship to published SBSP studies

The report does not convert published scenarios into default model inputs. All 29 defaults are study-authored exploratory assumptions. External evidence provides technical context, range inspiration, definition checks or consistency comparisons only.

Study	Published cost context	How to read it alongside this model
UK SBSP Phase 2 (2021)	£35-79/MWh (p10-p90), with p50 about £50/MWh in 2018 prices for a 2040 commissioning case.	Architecture and future-scenario context. Different maturity, financing, price basis and cost boundary prevent direct comparison.
DESNZ small-scale SBSP study (published 2026)	2024 GBP: £335-595/MWh (2030), £154-249/MWh (2035), and £87-129/MWh (2040).	Current UK architecture and system-value context. Different scale, orbit, procurement and hurdle rates mean band overlap is not validation.
NASA OTPS assessment (2024)	FY2022 USD: $610/$1,590 per MWh baseline cases and $30/$80 per MWh under combined favourable assumptions.	Shows the importance of combined favourable assumptions. No NASA cost value is imported into this model.

The wide published range is itself evidence that SBSP cost cannot be discussed independently of architecture, maturity, finance and system boundary. This project's contribution is to expose continuous parameter thresholds, cost denominators and executable calculations, not to claim greater predictive accuracy than fixed-year architecture studies.

Validation, uncertainty and remaining limitations

Automated validation covers reciprocal specific-power/specific-mass conversion, the 2 GW mass regression, no second efficiency division, all six power stages, rated-cost assignment, launch-mode exclusivity, launch rounding, replacement bases, construction shares, DCF/CRF convergence, zero-rate handling, invalid inputs, Python/browser parity, bilingual numerical parity and full regeneration.

- No physical design of antenna area, frequency, beam geometry, sidelobes, power density, land, weather or safety constraints.
- No vehicle manifest, structural load, thermal, radiation, failure, spares, discrete replacement mission or detailed launch schedule.
- No tax, inflation, financing tranches, learning curves or architecture-parameter correlations.
- No GB dispatch, network reinforcement, balancing, capacity value or market-revenue model.
- Exploration bounds are not probability distributions; no P10/P50/P90 labels are produced, and favourable one-way bounds should not be treated as jointly most likely.

Generated from the executable model. Price year: 2024 real GBP. Valuation base: start of construction (t=0).

Conclusion and next steps

Within this study boundary, the reference result of **£86.03/MWh** sits inside the study-defined £80-120/MWh discussion region and below the £100/MWh screen. The model also shows that, with every other assumption held at its reference value, launch cost to staging orbit alone can cross the £80 or £60/MWh thresholds. That is a conditional sensitivity result, not a forecast of launch-vehicle maturity, project schedule, system safety or finance availability. The defensible conclusion is that SBSP has a quantifiable low-cost condition space; whether those conditions can coexist in one buildable architecture remains an engineering and finance question.

Priority next steps:

- Replace exploratory system specific mass with an architecture-specific bill of materials, structural margin and grid-delivered specific power.
- Build a discrete launch manifest and orbit-transfer/refuelling plan, then test whether the per-kg price represents a procurable service.
- Add physical and regulatory constraints for beam design, rectenna, land, safety, weather and grid connection.
- Model discount rate, delay, failure and replacement as correlated uncertainties to produce P10/P50/P90 results.

Appendix A: Full parameter register

This register preserves all 29 defaults, explored ranges, cost or power denominators and limitations for audit. The main findings should not be interpreted outside these boundaries.

Parameter	Reference	Explored range	Denominator / boundary	Limitation
Delivered grid capacity	2000 MW-delivered AC	500-5000	AC power at the grid connection point	Scale input; the simplified model is nearly scale-neutral.
Solar-to-DC conversion efficiency	0.35 fraction	0.2-0.45	Incident solar power to space DC-bus output	Architecture-dependent exploratory value; not an independently verified flight-system value.
DC-to-RF efficiency	0.7 fraction	0.4-0.85	Space DC-bus input to emitted RF power	Excludes propagation and rectenna conversion losses.
RF transmission efficiency	0.98 fraction	0.85-0.995	Emitted RF power to RF incident on the rectenna	Simplified beam-capture proxy; beam geometry sidelobes weather and safety constraints are not explicitly modelled.
Rectenna RF-to-DC efficiency	0.65 fraction	0.4-0.85	Incident RF power at rectenna to rectenna DC output	Exploratory system-average conversion proxy.
DC-to-grid AC efficiency	0.96 fraction	0.9-0.99	Rectenna DC output to AC at the grid connection point	Includes simplified inversion and local electrical losses only.
Delivered capacity factor	0.9 fraction	0.6-0.98	Annual AC energy divided by rated delivered AC capacity times 8760	Combines availability outages and operating constraints; not a measured forecast.
Operating lifetime after commissioning	30 years	15-45	Full operating years after commissioning	Engineering retirement and economic life are simplified into one duration.
Real project discount rate	0.065 fraction	0.03-0.2	Discounting from start of construction t=0	Not an observed SBSP WACC; tax financing tranches and inflation are excluded.
Construction duration	4 years	0-8	Years from valuation base to commissioning	Default uses equal annual spend shares; project-specific scheduling is not represented.
Annual delivered-output degradation	0.005 fraction/year	0-0.03	Year-on-year reduction in delivered AC energy after the first operating year	Simplified fleet-average degradation proxy.
System specific mass	5 kg/kW-delivered	0.5-10	Complete operational orbital hardware mass per kW delivered AC at the grid boundary	Delivered-power basis; never divide by efficiency again. Published values can be architecture-specific or manufacturer-claimed.
Space-generation hardware cost	0.4 GBP/W-DC	0.05-5	Space DC-bus rated output excluding RF transmitter hardware	Includes collection structure conversion and DC conditioning only; boundary is mutually exclusive with transmitter cost.
RF transmitter hardware cost	0.1 GBP/W-RF emitted	0.02-1	Emitted RF rated power excluding generation and DC-bus hardware	Includes only DC-to-RF and aperture hardware; no generation hardware is counted here.

Parameter	Reference	Explored range	Denominator / boundary	Limitation
Launch service cost to staging orbit	500 GBP/kg to staging orbit	20-5000	Operational orbital hardware mass delivered to the defined LEO staging boundary	Not a GEO delivery quotation; procurement cadence integration and destination materially affect price.
Launch price per flight	1e+08 GBP/flight	1e+07-3e+08	One mutually exclusive launch flight to the staging orbit	Used only when per-flight mode is active; never added to per-kg launch cost.
Effective payload per flight	100000 kg/flight	20000-250000	Maximum manifested hardware mass reaching the staging orbit per flight	Vehicle and orbit dependent; used for launch-count diagnostics in both modes.
Payload utilisation	0.85 fraction	0.5-1	Average used fraction of effective payload capability	Does not model packaging volume schedule or rideshare constraints.
Staging-to-operational-orbit transfer cost	100 GBP/kg final hardware	0-3000	Final operational hardware moved from staging orbit to operational orbit	Must cover transfer vehicle propellant refuelling missions operations and payload penalty; a low value does not prove cheap HEO delivery.
In-orbit assembly and deployment cost	200 GBP/kg operational hardware	0-3000	Operational orbital hardware assembled deployed and commissioned	Exploratory allowance; robotic operations inspection spares and commissioning are bundled.
Rectenna cost	0.15 GBP/W-delivered AC	0.02-1	Rated AC power delivered at the grid boundary	Bundled proxy; antenna area frequency beam geometry sidelobes power-density land weather and safety are not explicit.
Grid connection cost	100 GBP/kW-delivered AC	20-300	Rated AC power at the grid connection point	Simplified local connection allowance; wider transmission reinforcement is excluded.
Initial programme contingency	0.2 fraction of pre-contingency initial CAPEX	0-0.5	Applied once to pre-contingency initial construction CAPEX	Not applied recursively to annual O&M or replacement allowances.
Space-hardware replacement rate	0.006 fraction/year	0-0.03	Fraction of eligible space hardware plus associated launch transfer and assembly cost replaced each year	Simplified annual equivalent rather than discrete failure and logistics scheduling.
Ground-hardware replacement rate	0.003 fraction/year	0-0.02	Fraction of eligible rectenna and grid cost replaced each year	Does not model component-specific replacement intervals.
Fixed O&M rate	0.01 fraction/year	0.002-0.05	Applied to generation transmitter rectenna and grid assets only	Excludes contingency initial launch transfer and assembly from its base.
Variable O&M	3 GBP/MWh delivered	0-20	Each MWh of AC energy delivered at the grid boundary	Exploratory allowance; market charges and downstream system costs are excluded.
Terminal decommissioning cost	0.02 fraction of initial CAPEX	0-0.2	Applied at the end of operating life to initial CAPEX	Timing and disposal obligations are simplified.
Terminal residual value	0 fraction of initial CAPEX	0-0.2	Credit at end of operating life relative to initial CAPEX	No default credit; salvage markets and liabilities are uncertain.

Appendix B: External evidence map

The table links parameters and claims to auditable external evidence, with document locators and comparability limits. Unless explicitly stated, published values do not replace model defaults.

Parameter / claim	Evidence role	External source and locator	Evidence context and comparability limit
Delivered grid capacity	external precedent	UK_SBSP_2021_PHASE2 · PDF pp.18-19	The 2021 architecture-specific case uses a 2 GW delivered plant. Numeric context: 2 GW CASSIOPeiA case. Comparability limit: The same number is used here as a study scale anchor, not as evidence that this architecture is preferred.
Operating lifetime	external precedent	UK_SBSP_2021_PHASE2 · PDF pp.18-19	The 2021 utility-scale economic case uses a 30-year operating life. Numeric context: 30 years; the later small-scale study uses 15 years. Comparability limit: Lifetime remains architecture-dependent and the model does not demonstrate survivability or refurbishment feasibility.
Computed end-to-end efficiency	technical context	CALTECH_SBSP_2022 · abstract	A lightweight modular concept reports substantial full-chain losses. Numeric context: 7-14% end-to-end efficiency. Comparability limit: Concept and subsystem boundaries differ; this is a consistency check on the computed output rather than a model input.
Computed end-to-end efficiency	source-inspired assumption	UK_SBSP_2021_ANNEX_B · PDF p.4	Products of reported component efficiencies provide context for a stage-resolved chain. Numeric context: approximately 14.85-24.39% from independent factor products. Comparability limit: Independent extrema do not define a probability distribution or a validated integrated design.
Computed end-to-end efficiency	external consistency check	DESNZ_SBSP_2025 · Table 1, PDF pp.11-12; reference-design selection p.13	The official small-scale study reports architecture-level total-efficiency cases. Numeric context: 11.7-19.9%; selected Space Solar small-scale reference design 14.9%. Comparability limit: Major values may be manufacturer-claimed or inferred and definitions may differ.
Delivered capacity factor	scope qualification	DESNZ_SBSP_2025 · Table 3, PDF p.16	The official study demonstrates that utilization varies materially by orbit and by the measured boundary. Numeric context: HEO: UK rectenna 95.7%, satellite average 53.9%; circular LEO: satellite average 20.1%, UK rectenna 21.5%. Comparability limit: Rectenna utilization is not equivalent to this model's delivered capacity factor; the 90% value remains a study assumption.
System specific mass	technical context	CALTECH_SBSP_2022 · abstract	Lightweight structures are central to proposed modular SBSP architectures. Numeric context: 160 g/m2 areal density. Comparability limit: Areal density cannot be converted to whole-system kg/kW-delivered without a complete delivered-power and subsystem boundary.
System specific mass	external consistency check	DESNZ_SBSP_2025 · Table 1, PDF pp.11-12; reference-design selection p.13	Architecture specific power is ground-delivered power per unit orbital mass; its reciprocal is kg/kW-delivered. Numeric context: 0.15-0.67 kW-delivered/kg; 0.67 converts to 1.4925 kg/kW-delivered. Comparability limit: Architecture-specific values are partly manufacturer-claimed; conversion does not verify engineering readiness and the model default remains exploratory.
Launch service cost to staging orbit	external range	UK_SBSP_2021_ANNEX_B · PDF p.5 and p.11	The 2021 input review uses a broad architecture-specific spacelift range. Numeric context: £358-2,410/kg. Comparability limit: Orbit procurement price year and service boundary differ from this staging-orbit variable.

Parameter / claim	Evidence role	External source and locator	Evidence context and comparability limit
Launch service cost to staging orbit	external comparison	DESNZ_SBSP_2025 · Table 12 p.37; Tables 27-28 pp.71-73	Launch cost is a dominant driver in the official small-scale cases. Numeric context: central HEO £1,647/1,400/1,153 per kg; optimistic £1,188/1,008/827 for 2030/35/40. Comparability limit: HEO delivery is not comparable to a staging-orbit service plus an exploratory transfer proxy.
Launch service cost to staging orbit	external precedent	NASA_OTPS_SBSP_2024 · PDF p.10	NASA's favourable combined case includes a low launch-price assumption. Numeric context: nominal $500/kg; $425/kg after a 15% quantity discount. Comparability limit: FY2022 USD US procurement and architecture boundaries differ; no direct input transfer is made.
Real project discount rate	financing context	DESNZ_SBSP_2025 · Table 12 p.37; Table 26 pp.68-69	Published SBSP cases use real pre-tax hurdle rates that decline with maturity and investor class. Numeric context: 20%/13.2%/9.06% scenario hurdles; investor cases 14.9% to 4.83%. Comparability limit: Hurdle rate is not identical to a project discount rate; 6.5% remains study-authored.
Real project discount rate	source-inspired assumption	DESNZ_EGC_2025_ANNEX_A · Technical and Cost Assumptions, row 38	A 6.5% hurdle-rate input appears for selected mature terrestrial generators. Numeric context: 6.5% for large solar and onshore wind. Comparability limit: This is not an observed SBSP rate and does not specify tax or financing structure.
Grid connection cost	source-inspired assumption	DESNZ_SBSP_2025 · PDF p.70	The small-scale study provides a direct grid-infrastructure allowance per MW. Numeric context: £0.15m/MW = £150/kW. Comparability limit: The model uses £100/kW as its own simplified allowance; scope and site conditions may differ.
Fixed O&M	source-inspired assumption	UK_SBSP_2021_ANNEX_B · Table B1, PDF p.4	Annex B uses a triangular O&M factor derived from terrestrial generation technologies. Numeric context: triangular lower/mode/upper values of 0.8%/1.9%/4.7%. Comparability limit: Definitions and cost bases vary; the model's 1% of CAPEX per year remains a study assumption.
Reference-case LCOE	external consistency check	DESNZ_SBSP_2025 · executive summary, PDF pp.3-4	The project anchor lies inside one published early small-scale band. Numeric context: 2024 GBP: £335-595/MWh (2030), £154-249 (2035), £87-129 (2040). Comparability limit: This is not validation: scale, architecture, orbit, scenario year, financing and accounting boundaries differ.
Published low-cost scenarios	external comparison	UK_SBSP_2021_PHASE2 · Executive summary p.3; assumptions pp.18-19; Figure 5 p.22 and Table 4 p.23	The earlier UK architecture study reports a much lower future case. Numeric context: £35-79/MWh (p10-p90), p50 about £50/MWh, 2018 prices; 2 GW; 30 years; 20% hurdle. Comparability limit: Architecture, learning, commissioning year, financing and price basis differ; values are not ranked as like-for-like forecasts.
Architecture dependence	external comparison	NASA_OTPS_SBSP_2024 · executive summary pp.3-4; pp.7-10	NASA's selected baseline and favourable-combination cases differ in lifecycle cost by roughly a factor of 20. Numeric context: FY2022 USD: $610/$1,590 per MWh baseline; $30/$80 per MWh favourable combination. Comparability limit: US architectures, service profiles, currency and lifecycle boundary differ; no NASA value is imported into this model.
Importance of launch cost	scope qualification	DESNZ_SBSP_2025 · Table 14, PDF p.38	The official small-scale study attributes most of its LCOE variance to launch assumptions. Numeric context: 55.5%/62.9%/64.0% for 2030/35/40. Comparability limit: This project may rank specific mass first within its own one-way ranges; neither ranking is universal.

Parameter / claim	Evidence role	External source and locator	Evidence context and comparability limit
Power-system value	system context	DESNZ_SBSP_2025 · executive summary, PDF pp.3-4	The official system cases reduce each reported LCOE by the same system-benefit adjustment. Numeric context: adjusting for system benefits reduces LCOE by £21/MWh in all cases. Comparability limit: The present model does not monetize whole-system value; £21/MWh is not subtracted from project LCOE.
System-adjusted comparison	benchmark definition	BEIS_EGC_2020 · Tables 7.1-7.3	Enhanced LCOE illustrates that wider system impacts can alter technology comparisons. Numeric context: transcribed ranges stored in uk_system_adjusted_costs.csv. Comparability limit: 2018 real GBP and scenario-specific; indicative only and not directly harmonised with the model's 2024 real GBP basis.

Appendix C: Methodological correction and version history

v1.x multiplied a 1.5 kg/kW value that should have been delivered-power-normalised by 'delivered power / efficiency', dividing by efficiency a second time. For 2 GW at 20% efficiency, the old expression gives **15,000,000 kg (15,000 tonnes)**; the corrected expression gives **3,000,000 kg (3,000 tonnes)**. With all other v2 comparison inputs identical, the erroneous mass boundary gives **£116.45/MWh**, versus **£43.45/MWh** on the corrected boundary. Historical v1.x thresholds are therefore not directly comparable with v2.0.

The DESNZ/Frazer-Nash report defines architecture specific power as ground-delivered power per orbital mass. The reciprocal of 0.67 kW-delivered/kg is **1.4925 kg/kW-delivered** (Table 1, PDF pp.11-12; reference-design selection p.13). Values remain architecture-specific and may include manufacturer claims that were not independently verified.

References

ASSUMPTION_THIS_STUDY is the project's internal assumption record, not an external publication. The bibliography lists only external sources actually used in the evidence map, deduplicated by source ID.

- **DESNZ_EGC_2025_ANNEX_A** - Department for Energy Security and Net Zero (2026-01-14). Annex A: Additional estimates and key assumptions 2025. Role in this report: UK generation benchmark input workbook. Accessed 2026-07-19.
- **BEIS_EGC_2020** - Department for Business, Energy and Industrial Strategy (2020-08-24). BEIS Electricity Generation Costs (2020). Role in this report: System-adjusted benchmark source. Accessed 2026-07-19.
- **UK_SBSP_2021_PHASE2** - Frazer-Nash Consultancy for BEIS (2021-04-23). Space Based Solar Power as a Contributor to Net Zero - Phase 2: Economic Feasibility. Role in this report: External SBSP economic study. Accessed 2026-07-19.
- **UK_SBSP_2021_ANNEX_B** - Frazer-Nash Consultancy for BEIS (2021-04-23). Space Based Solar Power as a Contributor to Net Zero - Phase 2: Economic Feasibility - Annex B: Input Data Sources. Role in this report: External SBSP input evidence. Accessed 2026-07-19.
- **DESNZ_SBSP_2025** - Frazer-Nash Consultancy; Space Solar Engineering Ltd; Imperial College London for DESNZ (2026-02-13). Feasibility of Small-Scale Space Based Solar Power (SBSP) Systems for Early Market Adoption. Role in this report: External SBSP study. Accessed 2026-07-19.
- **NASA_OTPS_SBSP_2024** - NASA Office of Technology, Policy, and Strategy (2024-01-11). Space Based Solar Power. Role in this report: External SBSP study. Accessed 2026-07-19.
- **CALTECH_SBSP_2022** - Caltech Space Solar Power Project (2022-06-16). A Lightweight Space-based Solar Power Generation and Transmission Satellite. Role in this report: SBSP technical context. Accessed 2026-07-19.